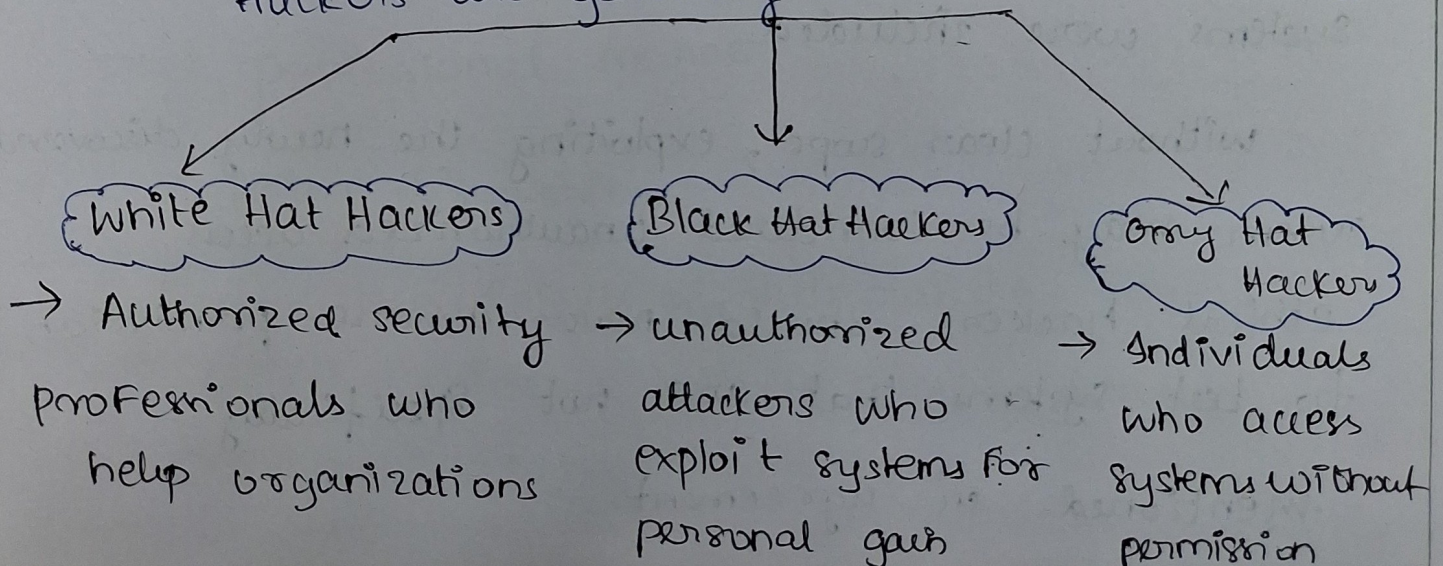


A company hires you to test its network but gives you only a one-line e-mail authorization with no defined scope. Halfway through testing, you discover a critical vulnerability in a system that wasn't mentioned in e-mail. do you exploit it to prove impact? Justify your answer using ethical Hacking principle.

Ethical Hacking is the process of identifying and testing security vulnerabilities in computer systems, networks, or applications with the permission of owner. The main objective is to improve cybersecurity by finding weakness before malicious hackers exploit them.

Hackers are generally classified into three types.



Would I Exploit Vulnerability ?

No, I would not exploit the vulnerability without obtaining additional written authorization from the company.

Although vulnerability is critical, exploiting it without permission would go beyond the authorized scope of testing. Instead I would document the vulnerability, report it immediately to the organization and request approval before performing any further testing.

Justification:-

① proper Authorization:-

The first principle of ethical hacking is obtaining proper authorization. In this case, the company only provided a one-line e-mail allowing network testing, but it did not clearly define which systems were included.

without clear scope, exploiting the newly discovered system may be considered unauthorized access.

ethical hackers should never assume permission to test systems that are not specifically mentioned. in agreement

② Respecting ethical principles

Ethical Hackers must follow professional ethics while performing security assessments.

These principles include

- * Acting honestly and responsibly
- * Avoiding unnecessary damage to systems
- * Maintaining the trust of organization.

③ Legal Framework.

Permission to perform security testing does not automatically allow unrestricted access to every system. Exploiting a vulnerability outside the agreed scope could be considered unauthorized activity and may result in legal consequences.

④ Responsible disclosure

Responsible disclosure means reporting discovered vulnerabilities to the organization in a secure and professional manner.

* After discovering :-

- Record the details of vulnerability
- collect evidence without causing damage.
- Inform company immediately

* explain the risk

* Request written permission before performing exploitation testing.

This approach helps the organization to fix problem

Avoid Annecessary Risk: -

A critical vulnerability can allow attackers to gain unauthorized access, steal sensitive information or disrupt business operations. However, exploiting it without permission may cause: -

* System crashes

* Data loss

* service interruptions

* Financial loss

* Organization reputation (damage)

Professional ethical hackers are expected to work with honesty, transparency and accountability

CONCLUSION:-

I would not exploit the vulnerability because it is outside my authorized scope. I would report it to the organization and seek written permission before conducting further testing.